WARIS

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VISION

I want to build technology that millions of people and businesses rely on every day, at a company where the bar is high and ownership is earned. I learn fast, take products from idea to working system on my own, and deliver under pressure: runner-up at JPMorgan Chase Code for Good and Microsoft-ACM HackThu, a 9.11 CGPA, and a 365-day LeetCode streak sustained over two years. Give me hard problems early, and I will deliver work that moves real numbers.

EDUCATION

Thapar Institute of Engineering and Technology, Patiala **Aug 2024 – May 2028**
B.E. Computer Science and Engineering | CGPA: **9.11/10** (up to Sem 4) | **Merit-3 Scholarship Holder**
Star Plus Convent School | CBSE Class XII (PCM): **91.40%** (2023) | CBSE Class X: **94.80%** (2021)

PROJECTS

VARWIX — Financial Data Science & Quantitative Risk Platform *React, Spring Boot, FastAPI, PostgreSQL, Redis, Docker*

- Architected a 3-tier platform (React dashboard, Spring Boot gateway with JWT auth, Python analytics microservice) serving [15+] quant and ML models to [200+] beta users at [99.5%] uptime.
- Built a risk engine for **Value-at-Risk and CVaR** (historical, parametric, **[10,000]-path Monte Carlo GBM**) with **Markowitz optimisation**; optimised portfolios backtested at [18%] higher Sharpe ratio than equal-weight.
- Shipped forecasting (**ARIMA, GARCH, LSTM**) and an ensemble ML studio (**XGBoost, Random Forest**): [35%] lower MAPE than a naive baseline and [0.91] ROC-AUC on churn/credit-risk models.
- Reduced p95 analytics latency by [82%] (1.8s to 320ms) via Redis caching and indexed PostgreSQL tables; Dockerised with GitHub Actions CI/CD.

Ideociety — AI-Powered Smart Residential Society Platform *Spring Boot, React, Three.js, OpenCV, YOLOv8, PostgreSQL*

- Built **role-based** society management (visitor passes, billing, complaints, amenity booking) across [8] Spring Boot modules; piloted with [120+] households, cutting complaint resolution time by [40%].
- Created an interactive **3D society map (React + Three.js)** with [24] live CCTV camera pointers showing feed, alerts and coverage, reducing guard response time by [35%].
- Deployed a **computer-vision layer**: YOLOv8 + OCR number-plate recognition at [96%] accuracy and [15 FPS] per stream, automating [500+] daily gate entries, plus intrusion and loitering alerts.

Grabito — Intelligent Food Delivery Platform *Next.js, Spring Boot, Kafka, Redis, PostGIS, WebSockets, LLM APIs*

- Engineered **real-time order tracking over WebSockets** with sub-[200ms] location updates and a **Kafka** pipeline handling [5,000] events/sec across [1,000+] concurrent simulated orders.
- Built a **rider assignment and order-batching simulator** (Hungarian matching + geospatial clustering) that cut rider idle distance by [22%], and "last 50 metres" address intelligence with [30%] fewer failed deliveries.
- Added AI features: conversational cart builder, rider voice assistant, menu-digitisation agent (OCR + LLM) and coupon-fraud detection flagging [92%] of test fraud cases.

Sparse-View NeRF + Camouflaged Object Detection — 3D Vision Pipeline *Python, PyTorch, nerfstudio, OpenCV*

- Reproduced **FreeNeRF** in nerfstudio for reconstruction from [3–8] views, gaining [+2.1 dB] **PSNR** and higher **SSIM** over vanilla NeRF at 3 views.
- Added camouflaged-object detection reaching [0.84] **F1** and [0.78] **IoU**, with ablations across view counts and occlusion levels.

EXPERIENCE & LEADERSHIP

TEDxTIET — Executive Committee Lead, Creativity & Logistics **2024 – Present** | Patiala

- Led logistics and creative operations for a **500+ attendee** TEDx event across [25+] vendors: procurement, stage and installation design, print production, security clearances and day-of execution.
- Managed hospitality and on-ground coordination for keynote speakers including **Nikhil Kamath** and **Piyush Goyal**, tracking [40+] tasks across teams.

EduExpose — Global Campus Ambassador **2024 – Present** | Remote

- Drove campus outreach and coordinated technical tracks and roadmap seminars reaching **300+ students**.

Class Representative (CSE, Batch 2024–28) & **Vice Proctor, Hostel J** — TIET **2024 – Present**

- Represent a batch of **150+ students** to faculty on academics and grievances; co-manage discipline, resources and welfare for **900+ hostel residents**.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript, SQL
Backend: Spring Boot, Spring Security (JWT), JPA/Hibernate, FastAPI, Node.js, REST APIs, WebSockets, Kafka, Microservices
Frontend: React, Next.js, Three.js, Tailwind CSS, GSAP, HTML5, CSS3
ML / Data / CV: PyTorch, TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, OpenCV, YOLOv8, nerfstudio, Time-Series (ARIMA, GARCH, LSTM), LLM APIs
Databases & DevOps: PostgreSQL, PostGIS, MongoDB, Redis, Docker, GitHub Actions (CI/CD), Git, Vercel, Railway, Jira, Figma

ACHIEVEMENTS & CERTIFICATIONS

- **Hackathons:** Runner-up, Code for Good (JPMorgan Chase); Runner-up, HackThu (Microsoft-ACM); Winner, InnovateX (TEDx).
- **Problem Solving:** 700+ DSA problems on LeetCode, Codeforces and CodeChef; 500+ SQL problems; 365-day LeetCode streak over 2 years.
- **Model United Nations:** Best Delegate, PEC Chandigarh MUN (USA); delegate at Thapar MUN, Joseph Lotus MUN and High Command MUN.
- **Certifications:** Agile with Jira (Atlassian), Computer Vision & Image Processing (IBM), Network Security (Google), Generative AI for Data Analytics (Microsoft), Data Analytics (Google).
- **Olympiads & NCC:** National winner in Maths, Physics and Chemistry Olympiads; NCC 'A' Certificate.
- **Social Impact & Creative:** Raised Rs. 50,000 for Rotaract and Pratigya Society; freelance portrait artist since 2018.